DIPLOMA PROJECT – FIRST PRE-DEFENSE REPORT

Project Title: Development of an Automated System for Monitoring and Controlling the Implementation of the AITU Development Strategy

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1. Project Proposal

1.1 Problem Statement

Astana IT University has adopted its Development Strategy for 2026-2030, which includes strategic goals, objectives, and key performance indicators (KPIs) with annual targets.

According to the initial supervisor consultation, AITU currently performs Development Strategy monitoring manually. This includes collecting information from structural departments, calculating the implementation status of tasks and key activities, and preparing reports. The absence of an automated monitoring system provides a practical motivation for the proposed project.

The project aims to address these challenges by developing a centralized automated system that organizes strategic information, tracks KPI performance, and provides up-to-date reports and visualizations.

1.2 Target Users

The system is intended for:

- **Strategy Department Staff:** Monitor strategic goals, track KPI results, and prepare reports.
- **Department Representatives:** Enter and update information about assigned indicators.
- **University Management:** View overall progress and performance reports.
- **System Administrator:** Manage users and access permissions.

The exact user roles and permissions will be clarified during requirements analysis.

1.3 Project Aim

To develop a centralized system that automates the monitoring and control of the AITU Development Strategy for 2026-2030 through KPI tracking, progress analysis, and reporting.

1.4 Project Objectives

1. Analyze the AITU Development Strategy and identify system requirements.
2. Review existing solutions and select suitable technologies.
3. Design the system architecture, database, and user interface.
4. Develop a centralized automated system for managing strategic goals, KPIs, annual targets, and actual results.
5. Implement automated progress calculations, dashboards, and reporting features.
6. Implement user authentication, access control, and data validation.
7. Test and evaluate the system's functionality, accuracy, and usability.

1.5 Expected Outcomes

The expected result is a functional automated system that allows authorized users to:

- Manage strategic goals, KPIs, and responsible departments.
- Enter and update actual KPI results.
- Compare planned and actual performance by year.
- Monitor progress through dashboards and reports.
- Access information according to assigned user roles.

The project will also include technical documentation, system diagrams, and testing results.

1.6 Project Scope

In Scope:

- Management of strategic goals and KPI indicators.
- Annual target and actual value tracking for 2026-2030.
- Automated KPI calculations and progress visualization.
- User authentication and role-based access control.
- Reporting and data validation.
- Basic tracking of monitoring data changes, if feasible within the final development schedule.

Out of Scope:

- Integration with internal AITU systems without authorized access.
- AI-based forecasting and decision-making.
- Replacement of existing university management systems.

The project will focus on an independently testable functional system for the diploma. Live integration or deployment at AITU is not part of the committed scope and would require separate authorization.

1.7 Engineering Complexity

The project covers the main stages of software development, including requirements analysis, database design, backend and frontend development, and testing.

The main technical challenges include managing related strategic data, processing different KPI types, implementing role-based access control, ensuring calculation accuracy, and generating reliable reports.

The initial implementation is limited to essential monitoring workflows suitable for one developer. Complex approval chains, predictive analytics, and institution-wide deployment are excluded from the initial scope.

Scope clarification: In this project, “control” initially means identifying reported versus missing results, displaying target–actual differences where appropriate, and restricting data modification according to assigned permissions. It does not imply an automated approval or enforcement process. This interpretation requires stakeholder confirmation.

2. Evidence of Relevance

2.1 Analysis of the AITU Development Strategy

The AITU Development Strategy for 2026-2030 includes strategic goals, objectives, and key performance indicators (KPIs) with annual target values.

The strategy covers several areas of university development, including education, research, international cooperation, digital transformation, and corporate governance.

Monitoring these indicators requires collecting actual results from university departments, comparing them with planned targets, determining the implementation status of strategic objectives and key activities, and preparing progress reports.

The approved strategy provides an initial foundation for the proposed system, including strategic directions, objectives, performance indicators, and annual targets.

2.2 Current Monitoring Process

The current monitoring process was clarified through written communication with the diploma project supervisor, Aliya Katyetova.

According to the supervisor, AITU does not currently have an automated system for monitoring the implementation of its Development Strategy.

The main activities are performed manually, including:

- Collecting information from university structural departments.
- Calculating the implementation status of tasks and key activities.
- Preparing reports on the implementation of the Development Strategy.

These findings confirm that the proposed project addresses an existing institutional process rather than a purely hypothetical problem.

Manual processing creates a potential need for a more structured and centralized approach to collecting information, calculating implementation status, and preparing reports. However, the exact time costs, frequency of reporting, and specific difficulties experienced by staff have not yet been measured.

Therefore, the project focuses on investigating how selected monitoring activities could be supported or automated through an information system.

2.3 Stakeholder Consultation

Consultation period: October 2026, before the first pre-defense.

Participant: Aliya Katyetova, Diploma Project Supervisor.

Consultation method: Written communication.

Purpose: To confirm the relevance of the proposed system, clarify the existing monitoring process, and obtain initial feedback on the project direction.

Main Findings

The consultation provided the following information:

1. AITU currently has no automated system for monitoring the implementation of its Development Strategy.
2. Information is collected manually from the university's structural departments.
3. The implementation status of tasks and key activities is calculated manually.
4. Monitoring reports are also prepared manually.
5. The proposed functionality is suitable as an initial project direction, but the final implementation approach has not yet been determined.
6. The university is considering Power BI or another technical solution for the digitalization of the monitoring process.

The supervisor also informed the student that a technical specification for the planned digitalization of the strategy monitoring process is being prepared for submission to the university's IT Department.

Requirements and Needs Identified

The consultation confirmed the existence of manual monitoring activities that may benefit from automation, particularly information collection, implementation-status calculation, and reporting.

These findings support further investigation of the proposed functions, including centralized data management, monitoring calculations, and report generation.

However, the detailed functional requirements, access permissions, reporting formats, and final technology selection have not yet been formally approved.

Questions Requiring Further Clarification

Further consultation is needed to determine:

- The specific responsibilities of each user group.
- How information is submitted and updated by structural departments.
- The rules used to calculate the implementation status of tasks and KPIs.

- The required reporting formats and frequency.
- Whether Power BI or another platform will be preferred.
- Whether university data or existing information systems can be used during development.

Consultation Evidence

The supervisor's written response provides preliminary qualitative evidence of the current monitoring process and the relevance of the proposed project.

This consultation is not presented as a completed staff survey or a comprehensive analysis of all stakeholders. Additional discussions may be needed during later development stages.

2.4 Initial Findings and Project Implications

The analysis of the approved AITU Development Strategy and the supervisor's feedback indicates that the project has practical relevance.

The following preliminary functions are proposed:

- Managing strategic goals, objectives, and KPIs.
- Recording planned and actual values for monitoring indicators.
- Organizing information received from responsible departments.
- Supporting the calculation and visualization of implementation progress.
- Identifying missing or incomplete monitoring information.
- Presenting monitoring results through dashboards and reports.

The supervisor confirmed that the proposed functionality is appropriate for the initial development stage. However, the final architecture and technology choices remain open.

The existing interactive prototype and system diagrams should therefore be considered preliminary design artifacts rather than approved specifications for institutional deployment.

The planned technical specification and further stakeholder consultations will help determine which functions should be implemented in the final version of the diploma project.

2.5 Conclusion

The relevance of the proposed system is supported by two sources: the approved AITU Development Strategy for 2026–2030 and the initial consultation with the diploma project supervisor.

The Development Strategy defines what needs to be monitored, while the supervisor's feedback confirms that information collection, implementation-status calculations, and reporting are currently performed manually.

This establishes a practical motivation for exploring an automated monitoring solution.

The consultation also confirms that the technical approach has not yet been finalized. The next stage will involve reviewing the university's planned technical specification, refining functional requirements, and evaluating possible implementation technologies.

2.6 References

Astana IT University. (2025). *Strategy of Advanced Development of Astana IT University for 2026–2030*. Approved December 17, 2025.

Katyetova, A. (2026, October). Personal communication regarding the current AITU strategy monitoring process and initial project requirements.

3. Literature Review

3.1 Introduction

Monitoring the implementation of a university development strategy requires more than collecting performance indicators. It also involves organizing data, comparing planned and actual results, and presenting information that supports management decisions.

This literature review examines research on performance management in higher education, KPI monitoring, business intelligence, dashboards, and information system development. The purpose is to identify approaches that can be applied to the proposed AITU strategy monitoring system.

3.2 Strategic Performance Management in Higher Education

The Balanced Scorecard (BSC) is one approach used to connect organizational objectives with measurable performance indicators.

Camilleri (2021) examined the use of the Balanced Scorecard in higher education and discussed the strengths and weaknesses of financial and non-financial measures. Taylor and Baines (2012) also studied the implementation of performance management in universities, focusing on how the Balanced Scorecard supports strategy monitoring and evaluation.

Al Jardali et al. (2021) developed a customizable Balanced Scorecard template for private universities in Lebanon, informed by comparative model analysis and semi-structured interviews. The need for adaptation to a specific institution is relevant to AITU.

Coskun and Nizaeva (2023) reviewed research on the Balanced Scorecard in educational institutions and identified trends in its use for strategic performance management.

These studies show that performance monitoring should reflect an institution's own strategy. For the AITU project, this means that the system should follow the approved strategic objectives and KPIs rather than introduce an unrelated set of indicators.

A limitation for this diploma is that management frameworks identify what should be measured but do not specify the local application workflow, roles, data constraints, or implementation rules for AITU. Those details require requirements engineering.

3.3 Business Intelligence and KPI Monitoring

Business intelligence (BI) can help institutions organize information and support data-based decision-making.

Sequeira et al. (2024) conducted a systematic literature review on the implementation of BI systems in higher education. Their study discusses institutional processes, strategic objectives, and BI architectures. It also identifies the need for a more systematic approach to BI implementation.

Stojanov and Daniel (2024) reviewed research on big data and analytics in higher education. Their work provides a broader view of how universities use institutional data for decision-making.

Sorour and Atkins (2024) presented an approach to quality monitoring in higher education using BI dashboards. Their study demonstrates how information from different sources can be presented in a centralized analytical interface.

Adot et al. (2023) proposed the SMART-QUAL indicator framework, which organizes and prioritizes measures for higher-education quality assessment. It supports the need for an explicit indicator structure, although AITU will use its approved KPI set.

These studies are relevant because the AITU strategy contains different types of KPIs with annual targets. The proposed system must organize these indicators consistently and allow users to compare planned and actual results.

BI and dashboard studies demonstrate the value of organized institutional data, but they do not remove the need to define reliable data entry, validation, role permissions, and institution-specific calculation rules. The proposed system must address these engineering tasks separately.

3.4 Dashboard Design and Usability

Dashboards are useful for presenting performance data, but their effectiveness depends on how the information is displayed.

Hjelle et al. (2024) conducted an experimental study on dashboard visualizations and organizational decision-making. Their findings indicate that the format, currency, and completeness of information can influence decision-making quality.

Almasi et al. (2023) reviewed methods used to evaluate dashboard usability. Although their review focuses on healthcare applications, its discussion of usability evaluation is relevant to the assessment of analytical interfaces in other fields.

For AITU, the dashboard should support a clear overview by year and strategic direction and distinguish missing results from recorded values. A visual ratio must not be interpreted as KPI achievement without an approved indicator-specific rule.

The usability of these functions should be evaluated during system testing rather than assumed from the interface design alone.

3.5 Information System Design and Evaluation

The development of an automated monitoring system also requires a structured software engineering process.

Hevner et al. (2004) described design science research as an approach in which information systems are developed and evaluated to address practical problems.

Peffers et al. (2007) presented a design science research methodology that includes problem identification, definition of objectives, design and development, demonstration, evaluation, and communication.

These approaches are relevant to the diploma project because the expected outcome is a working software artifact.The proposed system will be designed based on identified requirements and evaluated through

functional testing, KPI calculation checks, access control tests, and selected usability measures.

3.6 Structured Analysis of Selected Publications

Table 3.1 summarizes the 12 scientific publications used in this review. “Limitation for AITU” describes limits of direct applicability to this project, not necessarily deficiencies claimed by the original authors.

Study	**Approach / context**	**Relevant finding**	**Limitation for AITU**	**Design implication**
Camilleri (2021)	Critical literature review + staff interviews on BSC	Financial and non-financial measures involve trade-offs	Does not provide AITU data workflow	Keep indicator definitions aligned with institution
Taylor & Baines (2012)	Qualitative study of BSC use at four UK universities	Implementation context affects strategy measurement	Older; specific UK university context	Validate monitoring procedures locally
Al Jardali et al. (2021)	BSC template + semi-structured interviews in Lebanon	A customizable performance model can support universities	Different institutional setting; not deployable AITU software	Adapt rather than copy indicator structure
Coskun & Nizaeva (2023)	Bibliometric analysis of BSC in education	Documents BSC research trends and patterns	Research mapping, not software evaluation	Use as strategic management background
Sequeira et al. (2024)	Systematic literature review of BI in higher education	BI implementation needs structured planning	Roadmap not an AITU-specific operational system	Justify explicit data architecture

Study	Approach / context	Relevant finding	Limitation for AITU	Design implication
Stojanov & Daniel (2024)	Systematic review of HE big data/analytics	Outlines analytics applications and research issues	Focus wider than KPI entry and governance	Keep analytics scope deliberately limited
Sorour & Atkins (2024)	Literature-based QA framework and prototype dashboard	Demonstrates BI dashboard approach for QA monitoring	QA and big-data setting differs from AITU strategy	Use simple dashboards, avoid unnecessary big-data stack
Adot et al. (2023)	Institutional documents + interviews + literature; SMART-QUAL	Indicators can be grouped and prioritized systematically	Proposes its own indicator set, not AITU strategy targets	Represent KPI types and hierarchy explicitly
Hjelle et al. (2024)	Experiment using alternative dashboard visualizations	Information format, currency, completeness affect decisions	Decision task experiment; not KPI storage architecture	Emphasize understandable, complete visual displays
Almasi et al. (2023)	Systematic review of dashboard usability tools	Identifies criteria for assessing dashboard usability	Healthcare context; requires adapted tasks	Evaluate actual KPI task completion
Hevner et al. (2004)	Design science framework for information systems	Artifact building and evaluation address practical problems	Foundational method, not implementation instructions	Document design choices and demonstrate utility
Peffers et al. (2007)	Design science research methodology	Structures problem, design, demonstration, evaluation	Methodology does not define AITU business rules	Link requirements to tests and evaluation

Critical synthesis: Strategy management studies justify an institutional KPI hierarchy; BI research suggests data-driven reporting; dashboard studies emphasize clarity and usability; design-science literature requires an evaluated software artifact. None of these sources substitutes for validated AITU-specific data definitions, user permissions, and reporting procedures. Therefore, the intended contribution is a bounded implementation and evaluation of these workflows, not a claim to invent KPI monitoring.

3.7 Research Gap and Application to the Project

The reviewed studies provide useful approaches to strategic performance management, university KPIs, business intelligence, dashboard design, and software evaluation.

However, these approaches address different parts of the problem. Performance management research explains how institutions can measure progress, while BI studies focus on data organization and analysis. Dashboard research focuses on information presentation and usability.

For the AITU project, these aspects need to be considered together in a system based on the university's approved Development Strategy for 2026–2030.

The main design priorities identified from the literature are:

- A clear relationship between strategic goals, objectives, and KPIs.
- Consistent management of planned and actual values.
- Support for different KPI types.
- Reliable data processing and validation.
- Clear dashboards and reporting.
- Testing and evaluation of the developed system.

The project does not claim that no similar monitoring systems exist. Its contribution will be the design, implementation, and evaluation of a solution adapted to AITU's specific strategy and confirmed user requirements.

3.8 Conclusion

The literature shows that effective strategic monitoring depends on well-defined indicators, reliable data management, and clear reporting.

For the proposed AITU system, the reviewed approaches provide a basis for identifying requirements and making design decisions. The next stage is to compare existing software solutions and determine which technologies are most suitable for implementation.

References

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4. Existing System Comparison

4.1 Overview of Existing Systems

To understand existing approaches to strategic performance monitoring, four software solutions were reviewed: Microsoft Power BI, ClearPoint Strategy, Cascade, and Spider Impact.

These systems were selected because they provide features related to KPI tracking, performance analysis, strategic planning, or management reporting.

Microsoft Power BI is a business intelligence platform used to analyze data and create interactive reports. It supports data modeling, visualizations, and access controls. However, it is mainly an analytical tool, so a separate solution may be needed for managing data entry and approval workflows.

ClearPoint Strategy is designed for strategic planning and performance management. It connects organizational objectives, KPIs, responsible users, and reports within a single platform. Its structure is especially relevant to the proposed AITU system.

Cascade focuses on strategy execution. It connects strategic objectives with KPIs, initiatives, and organizational teams. It also provides dashboards, reports, and integrations with other systems.

Spider Impact is a performance management platform that supports scorecards, KPI tracking, strategy maps, dashboards, and reporting. It allows organizations to monitor actual results against targets and evaluate overall strategic performance.

4.2 Comparison Criteria

The systems were compared using eight criteria relevant to the AITU project:

1. Strategic goals and objectives management.
2. KPI target and actual value tracking.
3. Monitoring by time period.
4. Assignment of responsible users or departments.
5. Dashboards and data visualization.
6. Reporting capabilities.
7. User access and permissions.
8. Suitability for the AITU Development Strategy.

4.3 Comparative Analysis

Table 4.1. Comparison of existing strategy-monitoring solutions.

Criterion	Power BI	ClearPoint	Cascade	Spider Impact

Strategic goals management	Through data models	Supported	Supported	Supported
KPI target and actual tracking	Through data models	Supported	Supported	Supported
Time-based monitoring	Supported	Supported	Supported	Supported
Responsibility assignment	Requires external data or configuration	Supported	Supported	Supported
Dashboards	Supported	Supported	Supported	Supported
Reporting	Supported	Supported	Supported	Supported
Access permissions	Supported	Supported	Supported	Supported
Direct suitability for AITU	Requires customization	Requires customization	Requires customization	Requires customization

Note: Capability descriptions are based on vendors' publicly available product information (references below), not independent product trials. "Supported" does not imply the same workflow or configuration in every platform. Direct suitability for AITU cannot be established without its actual reporting and access requirements.

Comparison limitation: Power BI emphasizes analytical modeling and reporting, so KPI data-entry and governance workflows would normally need an additional data source or application. The strategy-oriented platforms include more built-in performance-management workflows but still require configuration for AITU's approved indicator hierarchy. No license-price comparison or institutional pilot has been performed.

4.4 Discussion

The comparison shows that existing software already provides many functions required for strategic performance monitoring.

ClearPoint, Cascade, and Spider Impact offer more complete strategy management features, while Power BI provides strong data analysis and visualization capabilities.

Therefore, the main challenge is not the absence of software for KPI monitoring. It is the need to implement a solution that follows AITU's Development Strategy for 2026–2030 and meets the requirements of its intended users.

The proposed system will focus on the university's strategic directions, objectives, annual KPI targets, responsible departments, and monitoring procedures.

It will also provide an opportunity to design and evaluate a complete software solution as part of an individual Software Engineering diploma project.

4.5 Conclusion

The reviewed systems demonstrate useful approaches to KPI management, strategy monitoring, dashboards, and reporting.

ClearPoint, Cascade, and Spider Impact provide relevant examples of strategic performance management, while Power BI is an important reference for analytical reporting.

The proposed AITU system will not attempt to reproduce all features of these commercial platforms. Instead, it will focus on the functions required to monitor the implementation of the university's approved strategy.

The final system architecture and technologies will be selected after requirements analysis and technology comparison.

4.6 References

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5. Technology Comparison

5.1 Introduction

This section compares technologies that could be used to develop the automated system for monitoring the AITU Development Strategy. Three technical decisions are considered: backend development, database management, and data visualization.

The comparison focuses on functionality, integration, development effort, and suitability for the project. The final technology selection will be confirmed after requirements analysis.

5.2 Backend Framework Comparison

Table 5.1. Backend framework alternatives.

Criteria	Django	FastAPI	Express.js
Language	Python	Python	JavaScript
Authentication	Built-in	Additional setup	Additional setup
Database access	Built-in ORM	External ORM	External libraries
Admin interface	Built-in	Custom	Custom
API support	Yes	Yes	Yes
Fit for MVP	Strong for integrated admin system	Strong for API-first system	Flexible; more assembly needed

Analysis: Django provides built-in authentication, database models, and an administrative interface. These features can reduce development effort for a system that manages users, KPIs, and related records. FastAPI is suitable for API-focused development, while Express.js provides flexibility with JavaScript.

Preliminary recommendation: Django for the first implementation, because its authentication, ORM, admin tools, and template rendering can reduce integration work for an individual developer. This is not a finalized institutional technology decision.

5.3 Database Comparison

Table 5.2. Relational database alternatives.

Criteria	PostgreSQL	MySQL	SQLite
Relational data	Yes	Yes	Yes
Data integrity constraints	Yes	Yes	Yes
Transactions	Yes	Yes	Yes
Concurrent write support	Yes	Yes	Limited
Setup complexity	Moderate	Moderate	Low
Suitability	High	High	Prototype

Analysis: The proposed system will store related information about strategic goals, objectives, indicators, annual targets, and responsible departments. A relational database is suitable for maintaining these relationships.

PostgreSQL and MySQL both support relational constraints and concurrent access. SQLite is easier to configure but less suitable for a system with multiple simultaneous data updates.

Preliminary recommendation: PostgreSQL for a multi-user final system, with SQLite permitted during local development or demonstration. The data model should avoid database-specific features where practical.

5.4 Data Visualization and Reporting Comparison

Table 5.3. Visualization and reporting alternatives.

Criteria	Power BI	Chart.js	Apache ECharts
Interactive dashboards	Yes	Yes	Yes
Built-in BI analysis	Yes	No	No
Custom chart design	Available	Flexible	Flexible

Integration	Requires configuration	JavaScript integration	JavaScript integration
Licensing	Depends on usage	Open source	Open source
Best use	Analytical reporting	Custom dashboards	Advanced charts

Analysis: Power BI provides tools for analytical reporting and interactive dashboards. However, sharing and embedding reports may require additional licenses and configuration.

Chart.js and Apache ECharts can be integrated into a custom system. They provide control over visualizations but require the developer to implement data processing and reporting logic.

Preliminary recommendation for a manageable MVP: Chart.js embedded in the application interface. Power BI remains an optional alternative for institutional analytics if its access, licensing, and integration requirements are confirmed.

5.5 Preliminary Technology Decisions

Table 5.4. Preliminary technology decisions.

Component	Preferred Option	Status
Backend	Django	Preliminary
Database	PostgreSQL (SQLite for local prototype)	Preliminary
Visualization	Chart.js (Power BI optional)	Preliminary; to validate

These choices are based on the initial project scope. They may change after consultation with stakeholders and a more detailed analysis of system requirements.

5.6 Conclusion

The comparison identified Django and PostgreSQL as suitable options for developing the system's core functionality and managing structured KPI data.

For the MVP, embedded charts may provide a simpler development path than a separate BI integration. Power BI remains under consideration only if stakeholder requirements justify the additional configuration and licensing implications.

The selected technologies should support a reliable, maintainable system without adding unnecessary complexity to the diploma project.

6. Requirements Specification

6.1 Overview

This section defines the initial functional and non-functional requirements for the automated system for monitoring the implementation of the AITU Development Strategy for 2026-2030.

The requirements are based on the approved strategy and the initial project scope. They will be reviewed and updated after discussions with the intended users.

The MoSCoW prioritization method is used to distinguish essential functions from additional features.

6.2 Functional Requirements

Functional requirements describe the operations that the system must perform.

Table 6.1. Functional requirements and acceptance criteria.

ID	Functional Requirement	Priority	Acceptance Criteria
FR-01	The system shall allow authorized users to sign in and sign out.	Must Have	A registered user can sign in with valid credentials. Invalid credentials are rejected.
FR-02	The system shall provide different access permissions based on user roles.	Must Have	A user without editing permission cannot modify KPI records.
FR-03	The system shall represent the approved strategic directions, objectives, and KPI definitions. Editing the structure shall be restricted to an authorized administrative role.	Must Have	The approved hierarchy can be loaded or created as structured records; valid relationships are retained. Unauthorized structural edits are rejected.
FR-04	The system shall store annual KPI targets for 2026–2030.	Must Have	Each KPI can have a separate target for each applicable year, including an explicitly missing target.
FR-05	The system shall allow authorized users to enter and update actual KPI results.	Must Have	A valid result can be saved and retrieved for the selected KPI and year.

FR-06	The system shall store numeric, percentage, and milestone/status KPI values while preserving their measurement types.	Must Have	Numeric, percentage and milestone values are stored and displayed according to their declared type; missing values remain distinct from zero.
FR-07	The system shall show target–actual differences for compatible numeric KPIs where a comparison rule has been defined.	Must Have	For numeric target 100 and actual 80, the difference is −20. No achievement or traffic-light status is inferred without an approved indicator rule.
FR-08	The system shall display KPI monitoring information by year and strategic direction.	Must Have	Selecting a year and direction shows only the matching indicators.
FR-09	The system shall associate KPIs with responsible departments or users.	Must Have	Each KPI can be linked to a responsible department and displays that assignment; detailed approval workflows are not required.
FR-10	The system shall provide a dashboard summarizing available KPI results.	Must Have	The dashboard lists relevant recorded and missing results by period; it does not imply unsupported aggregate achievement scores.
FR-11	The system shall generate a monitoring report for a selected period.	Must Have	For a selected year, an on-screen report lists KPI name, type, annual target, actual result, and supported comparison. File export is optional.
FR-12	The system shall validate information entered by users.	Must Have	Invalid values are rejected with an understandable message and are not saved.
FR-13	The system shall record changes made to KPI results.	Should Have	When implemented, an authorized reviewer can see the old value, new value, date/time, and account for an edited actual result.

FR-14	The system may provide notifications about missing or delayed KPI updates.	Could Have	If enabled, a notification is generated according to an agreed update deadline.

6.3 Non-Functional Requirements

Non-functional requirements define the expected quality of the system.

Table 6.2. Non-functional requirements and acceptance criteria.

ID	Non-Functional Requirement	Priority	Acceptance Criteria
NFR-01	The system shall protect access to restricted data.	Must Have	Unauthenticated access to protected functions is rejected, and permissions are checked for protected operations.
NFR-02	The system shall maintain data consistency.	Must Have	Invalid references are rejected and failed transactions do not leave incomplete records.
NFR-03	The system shall provide acceptable response times.	Should Have	Under a test load of 10 concurrent users and 500 KPI records, 95% of standard list and detail requests finish within 3 seconds.
NFR-04	The system shall provide a clear and consistent user interface.	Should Have	At least 4 out of 5 participants complete predefined basic tasks without assistance during prototype usability testing.
NFR-05	The system shall support data recovery during testing.	Should Have	A database backup can be restored successfully in a test environment without losing backed-up records.
NFR-06	The system shall be maintainable.	Must Have	The project includes setup instructions, configuration examples, and automated tests for core calculation rules.

NFR-07	The system shall support authorized access through its selected interface.	Must Have	All core workflows can be completed using the interface chosen during system design.

Performance and usability thresholds are provisional evaluation targets proposed for the diploma prototype, not requirements already approved by AITU.

6.4 Requirement Prioritization

The first working version will focus on the essential monitoring functions:

- Authentication and access control.
- Management of strategic goals and KPIs.
- Annual target and actual result tracking.
- KPI comparison and data validation.
- Dashboards and monitoring reports.

Responsible departments (FR-09) are included in the initial core workflow. Change history and backup recovery are secondary development priorities after the core workflow is stable.

Notifications, multi-stage approvals, external integrations, predictive analytics, and advanced cross-KPI scoring are excluded from the committed MVP; they may be explored only after stakeholder confirmation and feasibility review.

6.5 Preliminary KPI Data Model and Calculation Rules

The following conceptual entities reflect the approved strategy structure and the proposed MVP. The final schema will be specified after a workflow consultation. A document KPI code will not be assumed globally unique.

Entity	**Core fields (initial)**	**Purpose**
StrategicDirection	Internal ID, name	Groups strategic objectives
StrategicObjective	Internal ID, direction ID, title	Preserves the approved hierarchy
KPI	Internal ID, source code, objective ID, title, type, unit, department ID	Defines an indicator; internal ID is unique
AnnualTarget	KPI ID, year, target, missing flag	Keeps each annual plan separately
ActualResult	KPI ID, year, value/status, updated timestamp	Stores actual reporting result; supports missing

Entity	Core fields (initial)	Purpose
Department	Internal ID, name	Assigns accountable organizational unit
User / Role	Account, role, permissions	Restricts editing and viewing operations
ChangeLog (optional)	KPI/result ID, old/new values, timestamp, user	Traceability only if FR-13 is implemented

Table 6.3. Preliminary logical data entities (proposed; not yet stakeholder-approved).

Initial comparison rules (to validate): numeric results use Actual − Target; percentages use the same percentage-point difference when units match; milestones display the recorded state without arithmetic. Missing target or actual values are shown as “not available,” not zero. No direction-of-improvement or success classification is assumed until each relevant KPI rule is approved.

6.6 Requirements Validation

The initial requirements were derived from the AITU Development Strategy and the preliminary project scope.

Stakeholder consultation is still required to confirm:

- User roles and access permissions.
- Responsibility for entering and approving KPI results.
- Indicator-specific measurement types, missing-value conventions, and rules for interpreting “achievement”.
- Whether notifications, approvals, exports, or integrations are required at all.
- Notification and integration requirements.

Confirmed requirements will be updated before final implementation.

6.7 Conclusion

The initial requirements define a manageable scope for the proposed automated system. Functional requirements focus on KPI data management, progress monitoring, and reporting, while non-functional requirements address security, reliability, usability, and maintainability.

Each requirement includes a priority and a testable acceptance criterion. The specification will be refined after stakeholder feedback.

7. Architecture and Diagrams

7.1 System Architecture Overview

The proposed system will support the monitoring of the AITU Development Strategy for 2026-2030. It will manage strategic goals, KPIs, annual targets, actual results, and monitoring reports.

The initial architecture consists of three levels: user interaction, system functionality, and data storage. The final implementation technologies will be selected after the requirements are confirmed.

7.2 System Context Diagram

The System Context Diagram shows the system and its main users.

Four user groups are currently considered: Strategy Staff, Department Representatives, University Management, and System Administrators.

Strategy Staff manage KPI information and reports. Department Representatives provide actual KPI results. University Management reviews strategy implementation progress, while Administrators manage access permissions.

The published Development Strategy is an input document for manually preparing the initial records; this diagram does not represent an existing live connection to AITU internal information systems.

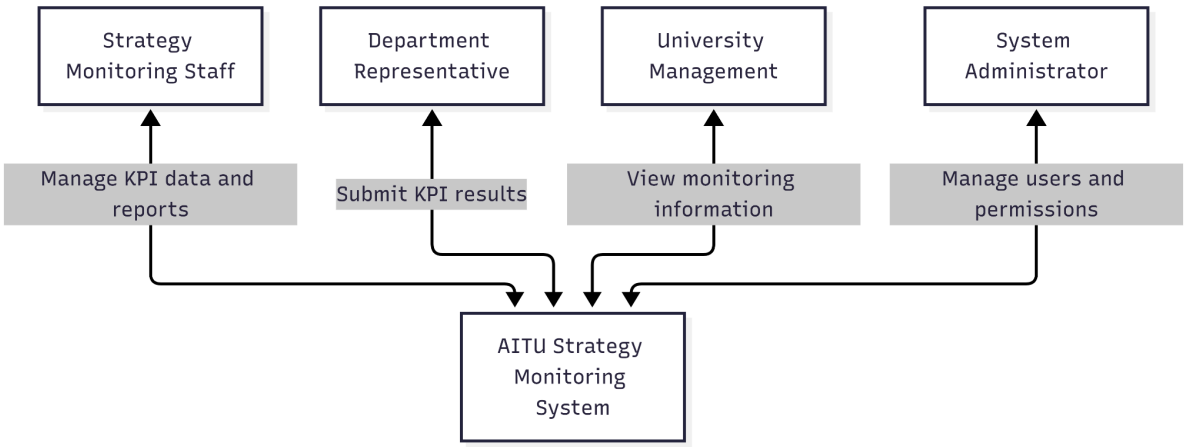


Figure 7.1. Proposed System Context Diagram (draft).

The roles and responsibilities shown in the diagram are preliminary and will be validated with stakeholders.

7.3 Use Case Diagram

The Use Case Diagram presents the main interactions between users and the system.

The key use cases include managing strategic goals and KPIs, entering actual results, viewing progress, generating reports, and managing user access.

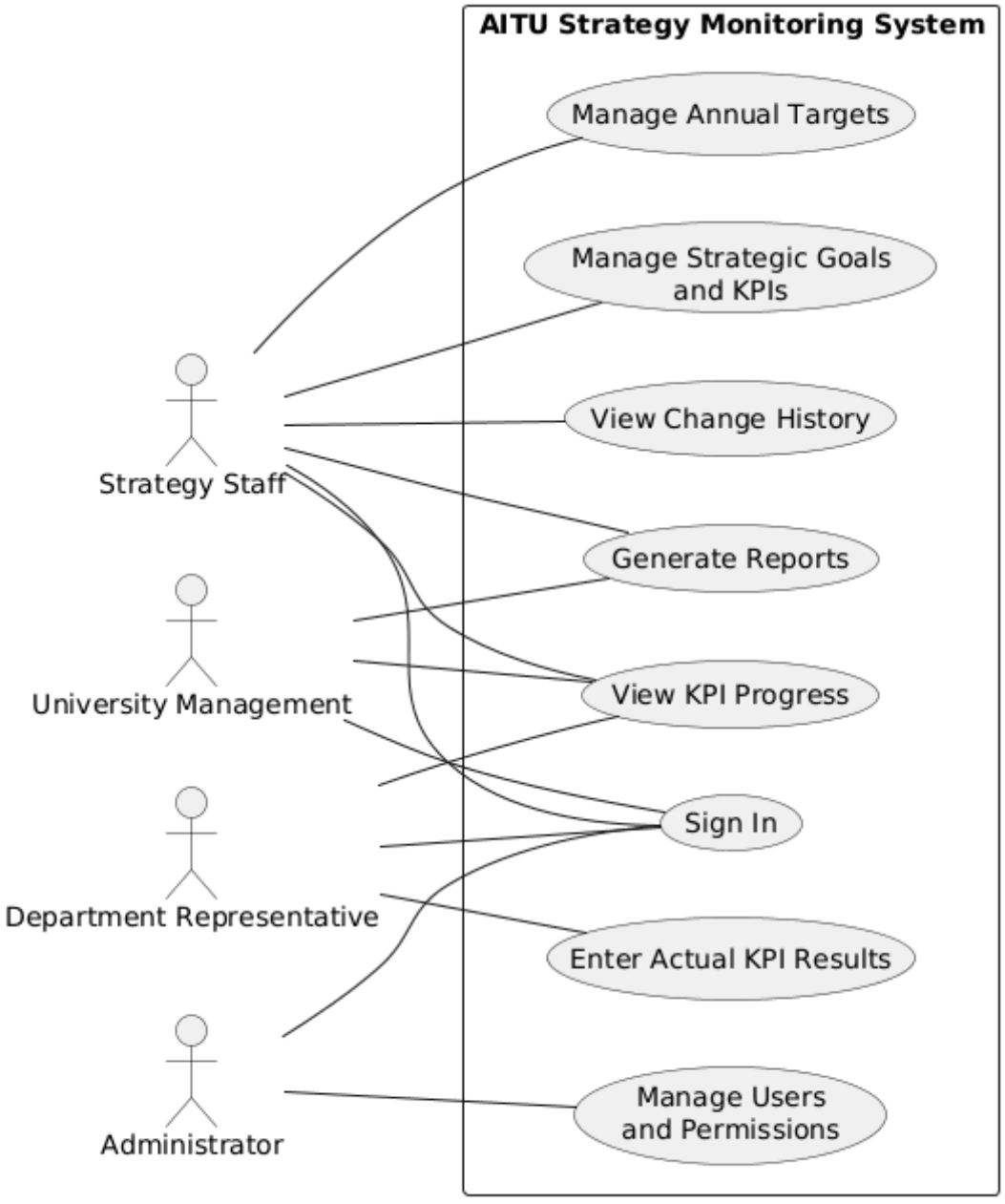


Figure 7.2. Use Case Diagram (proposed roles and scenarios).

The diagram is based on the initial functional requirements defined in Section 6.

7.4 Component Diagram

The draft Component Diagram shows the main functional parts of the system.

The User Interface allows users to access system functions. Authentication and Access Control manages permissions. Strategy and KPI Management handles strategic information and results.

KPI Calculation and Validation processes entered values, while Reporting and Dashboard presents monitoring information. Change History records modifications, and the Database stores the system data.

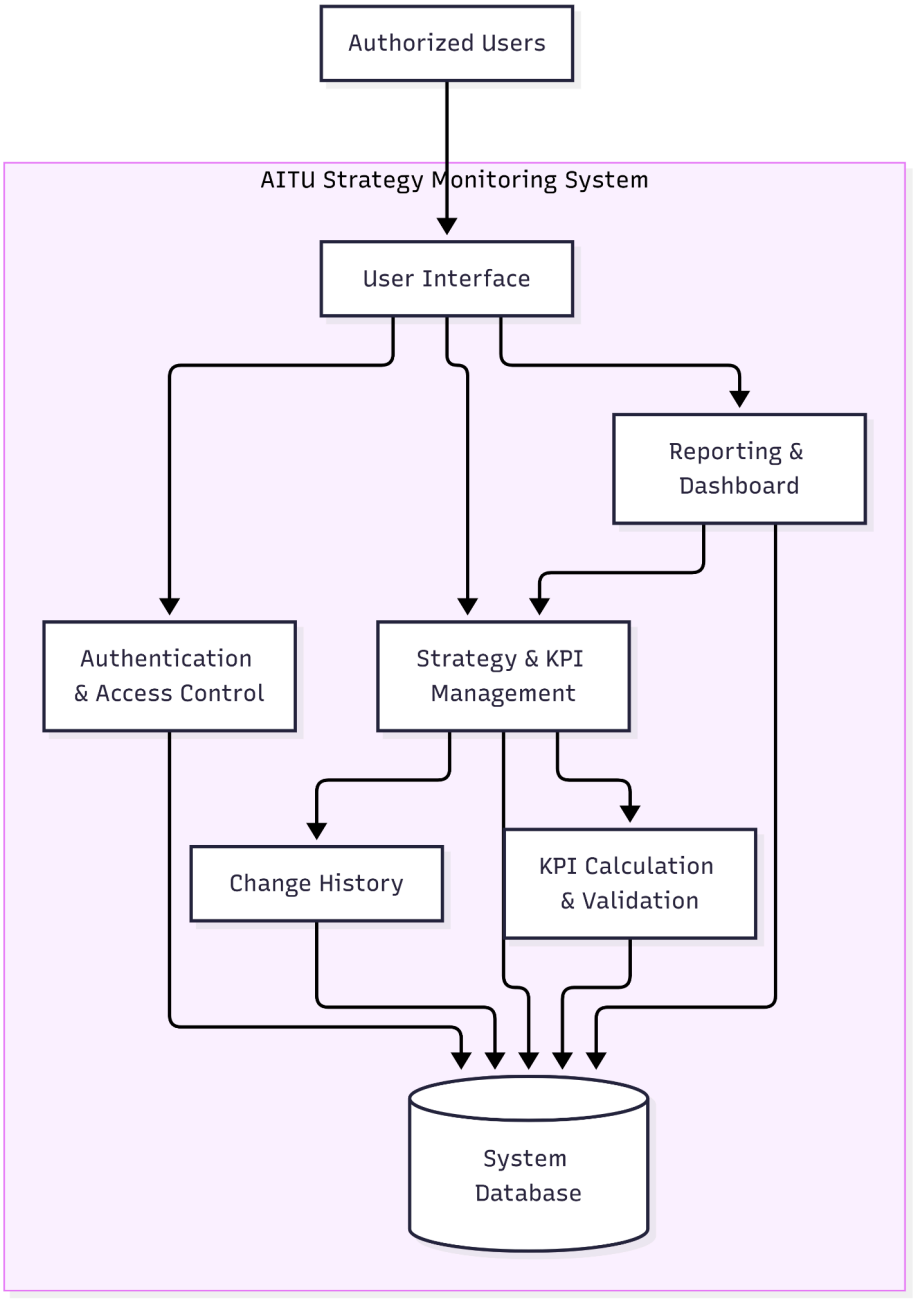


Figure 7.3. Draft Logical Component Diagram.

The diagram represents a preliminary logical architecture and does not depend on a specific backend framework or visualization technology.

7.5 Conclusion

The three diagrams describe the system boundaries, user interactions, and main software components.

They provide a starting point for the implementation of the proposed system. The architecture will be updated after stakeholder consultations and the final selection of technologies.

8. UX / Prototype

8.1 Prototype Overview

An interactive HTML prototype was prepared to demonstrate the main user interactions with the proposed AITU Strategy Monitoring System.

The prototype includes four main screens: Dashboard, KPI Management, KPI Details, and Reports. It demonstrates how users can view strategic indicators, monitor progress, update KPI results, and access reports.

The prototype uses demonstration data and does not include a production database or real user authentication. Its purpose is to illustrate the proposed functionality before full system development.

Prototype limitations: there is no real authentication, permission enforcement, persistent database, or institutional integration. KPI values, departmental assignments, and summary ratios on the screenshots are illustrative and must not be interpreted as university performance results.

8.2 Main Prototype Screens

Dashboard

The Dashboard provides an overview of strategy implementation. It displays KPI summaries, progress indicators, and monitoring information for the selected year.

Users can review general performance and navigate to more detailed KPI information.

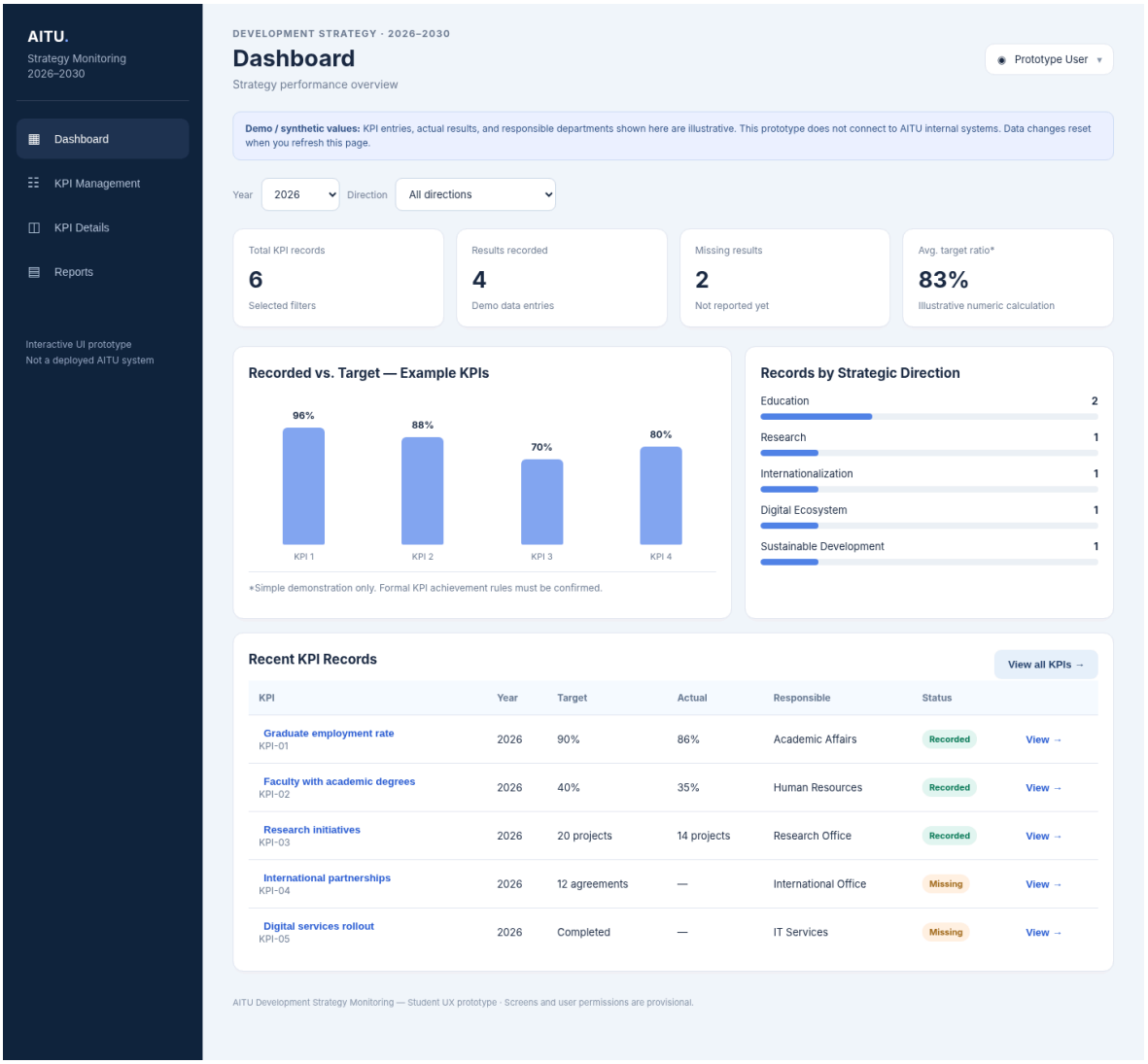


Figure 8.1. Dashboard – HTML prototype with synthetic data.

KPI Management

The KPI Management screen displays a list of strategic performance indicators.

Users can search for KPIs, filter records by year or strategic direction, and open individual indicators to view more information.

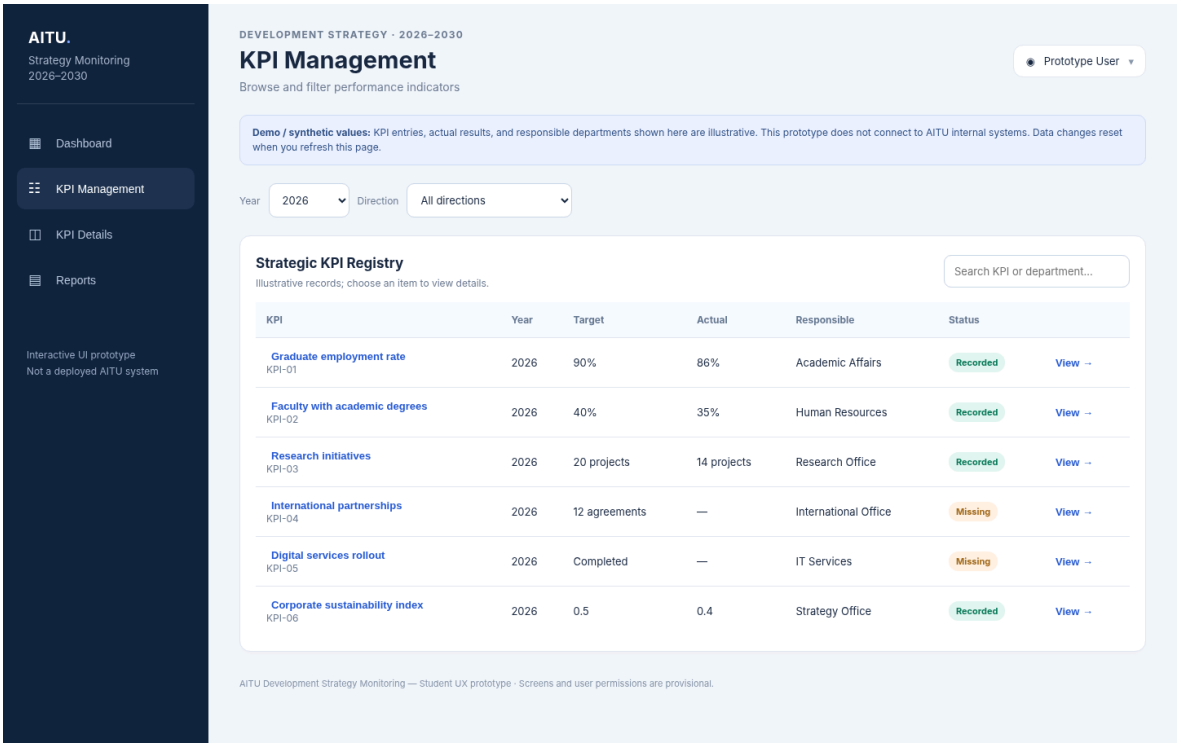


Figure 8.2. KPI Management – HTML prototype with synthetic data.

KPI Details

The KPI Details screen presents information about a selected indicator, including its target and actual values.

The prototype allows users to enter and update demonstration KPI results. This interaction illustrates the planned data management functionality.

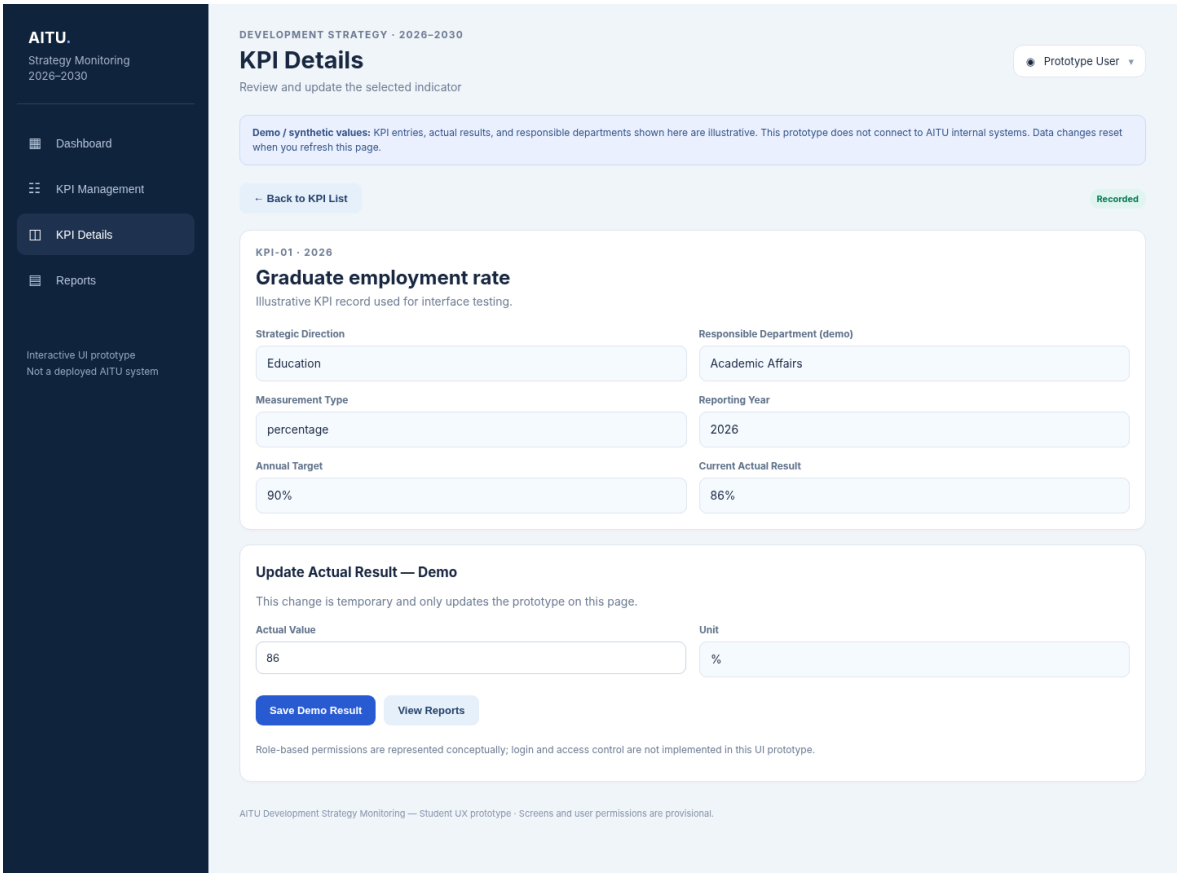


Figure 8.3. Details – HTML prototype with synthetic data.

Reports

The Reports screen demonstrates how monitoring information can be presented for a selected reporting period.

Users can select a year, generate a sample report, and review the available KPI results.

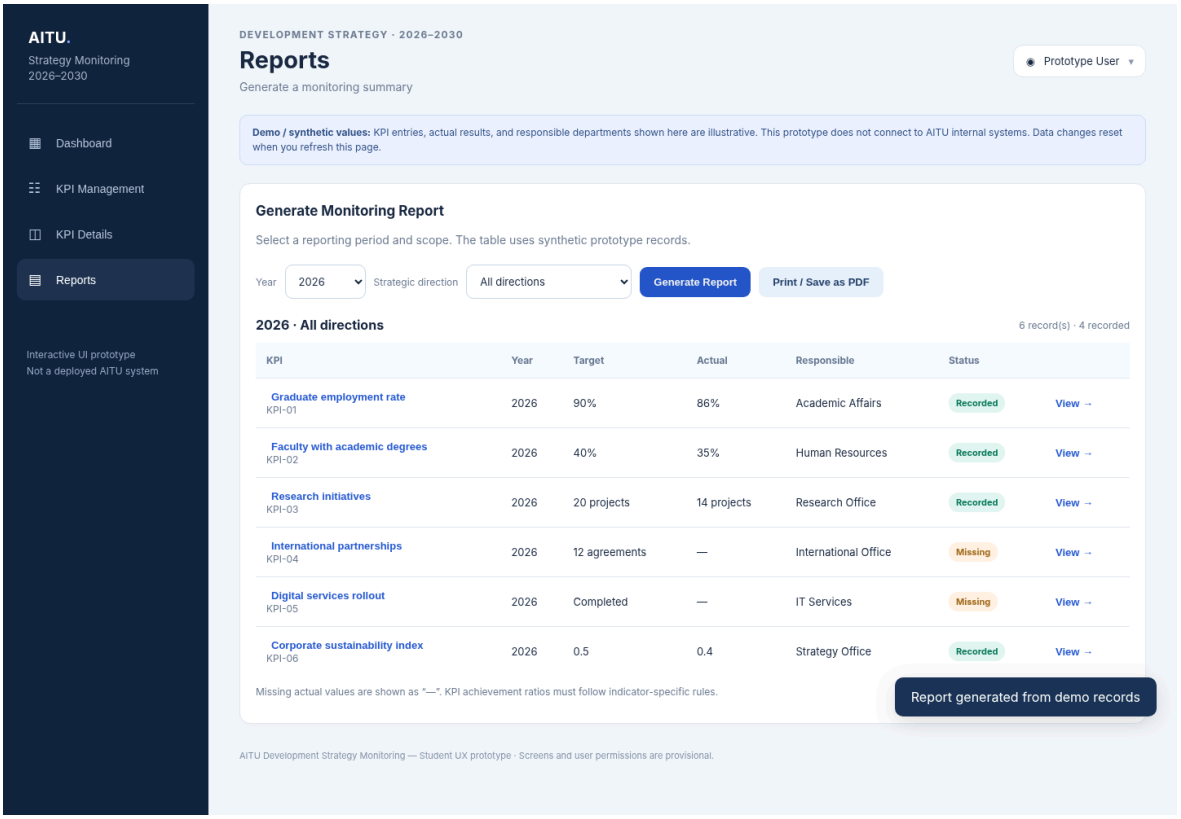


Figure 8.4. Reports – HTML prototype with synthetic data.

8.3 Main User Scenarios

Table 8.1. Prototype user scenarios.

ID	User Scenario	Expected Result
US-01	A user opens the Dashboard and selects a year.	KPI information for the selected year is displayed.
US-02	A user searches for a KPI in the KPI Management screen.	Relevant indicators are displayed.
US-03	A user opens a KPI and updates its demonstration actual value.	The updated value is displayed in the prototype.
US-04	A user selects a reporting period and generates a report.	A sample monitoring report is displayed.

8.4 Prototype Evaluation

The prototype illustrates KPI viewing, searching, temporary data edits, and example reporting. The screenshots use the initial synthetic 2026 dataset with all strategic directions selected; they are not evidence of formal functional or usability testing.

The interface will be reviewed during further development to evaluate navigation, clarity, and usability. Any changes will be based on stakeholder feedback and confirmed requirements.

8.5 Conclusion

The interactive prototype provides an initial representation of the proposed system and its key functions.

It helps demonstrate the main user workflows and provides a basis for further interface development. The final design will be refined after requirements validation.

9. Verification and Evaluation Plan

9.1 Verification Approach

The proposed AITU Strategy Monitoring System will be evaluated through functional, data validation, calculation, access control, performance, and usability testing.

The purpose of verification is to ensure that the system meets the functional and non-functional requirements defined in Section 6.

Testing will be conducted during implementation using synthetic data until access to authorized institutional data is available.

9.2 Test Plan

Table 9.1. Verification test cases (planned).

ID	Test Area	Test Scenario	Expected Result
TC-01	Authentication	Attempt to access a protected function without signing in.	Access is denied.
TC-02	Access Control	Attempt to modify KPI data without the required permission.	The modification is rejected.
TC-03	KPI Management	Create or edit a KPI with valid information.	The KPI is saved correctly.
TC-04	Data Entry	Enter a valid actual KPI value.	The value is stored and displayed correctly.
TC-05	Data Validation	Enter an invalid value for a numeric KPI.	The system displays a validation error.
TC-06	Calculation	Compare a numeric target of 100 with an actual value of 80.	The system displays a difference of -20.
TC-07	Missing Data	View a KPI without an annual target.	The missing target is not treated as zero.
TC-08	Filtering	Filter KPIs by year and strategic direction.	Only matching records are displayed.

TC-09	Reporting	Generate a report for a selected year.	The report contains the corresponding KPI results.
TC-10	Change History	Update an existing KPI result.	The modification is recorded in the change history.
TC-11	Percentage validation	Enter a percentage outside the accepted range for a percentage-type KPI.	The value is rejected with an explanatory message.
TC-12	Milestone KPI	Enter and retrieve a milestone/status result.	The status is shown without a numeric success ratio.
TC-13	Missing actual	Open a KPI without an actual result.	The result is labeled missing rather than represented as zero.
TC-14	Responsibility	Inspect a KPI with an assigned department.	The appropriate responsible unit is displayed.

9.3 Evaluation Metrics

Table 9.2. Evaluation metrics and provisional thresholds.

Metric	Measurement Method	Proposed Success Criterion
Functional Test Pass Rate	Passed tests / executed tests × 100%	100% of critical tests pass
Calculation Accuracy	Compare calculated results with independently prepared expected values	100% of defined calculation test cases pass
Data Validation	Execute valid and invalid input test cases	All defined validation cases pass
Access Control	Test actions under different user roles	No unauthorized actions succeed

Response Time	Measure common operations under the defined test workload	At least 95% complete within 3 seconds
Task Completion Rate	Observe users completing selected scenarios	At least 80% complete tasks without assistance

The performance target will initially be evaluated with approximately 500 KPI records and 10 concurrent users. These thresholds are preliminary engineering targets and will be reviewed after stakeholder consultation.

9.4 Evaluation Procedure

The evaluation will follow four stages.

Stage 1 - Test Data Preparation

Prepare synthetic records representing strategic directions, goals, KPIs, annual targets, and actual results for 2026–2030. Include numeric, percentage-based, milestone-based, and missing values.

Stage 2 - Functional Verification

Execute the defined test cases and compare actual results with expected outcomes. Record passed and failed tests and document identified defects.

Stage 3 - Performance and Security Verification

Measure response times under the planned workload and test access restrictions for different user roles.

Stage 4 - Usability Evaluation

Ask a small group of representative users to complete typical tasks, including finding a KPI, updating a result, and generating a report. Measure task completion and collect feedback on interface clarity.

9.5 Baseline and Comparison

A reproducible synthetic-data baseline will be prepared in a spreadsheet (e.g., Microsoft Excel), using the same KPI records and reporting tasks as the automated system. This allows a comparison even if access to an institutional baseline is not authorized.

For each task, the same operator will complete a prescribed workflow using the spreadsheet and the developed system. Task duration, completion, and errors will be recorded under comparable desktop conditions; order effects and familiarity will be noted. At least three repetitions per task will be attempted if feasible.

The median task time for spreadsheet and system conditions will be compared using percentage change: (T_baseline − T_system) / T_baseline × 100%. A provisional target is a 20% reduction for selected reporting tasks, but actual values will not be claimed until measured. An institutional workflow comparison remains optional, subject to permission.

9.6 Baseline Tasks and Measurement Conditions

The manual spreadsheet is a controlled comparison task, not an assertion that the university currently follows this exact Excel workflow. Only identical synthetic input records and task definitions will be compared.

Task	**Spreadsheet baseline**	**Proposed system**	**Recorded measure**
Find a KPI	Search a 2026 KPI among sample rows	Search/filter the KPI registry	Elapsed seconds, correctness
Update an actual result	Locate row, enter a value, verify it	Open KPI, edit result, confirm	Elapsed seconds, validation errors
Compile annual summary	Select rows and manually summarize available values	Generate the period report	Elapsed seconds, missing-value errors

9.7 Expected Evaluation Outcome

The evaluation is expected to determine whether the system correctly manages KPI information, performs calculations, protects data according to user permissions, and supports the main monitoring workflows.

Test results will be documented during implementation. The proposed metrics and thresholds may be adjusted after requirements validation.

10. Project Management and Risks

10.1 Project Management Approach

The development of the AITU Strategy Monitoring System will be organized using an iterative approach. The project will be divided into manageable tasks covering requirements analysis, system design, implementation, testing, and documentation.

GitHub will be used for version control, while YouTrack will be used to organize tasks, priorities, deadlines, and progress.

The project is being developed individually under the supervision of Aliya Katyetova.

10.2 Project Milestones

Table 10.1. Proposed project milestones.

Milestone	Main Activities	Expected Deliverable
M1 - First Pre-Defense	Research, requirements, comparisons, architecture, prototype, and testing plan	Initial project documentation and prototype
M2 – Requirements Finalization	Stakeholder consultation and revision of requirements	Validated requirements specification
M3 – Core Development	Database design, KPI management, data entry, and calculations	Initial functional system
M4 – Monitoring and Reporting	Dashboard, filters, reports, and access control	Integrated monitoring functionality
M5 – Testing and Refinement	Functional, security, performance, and usability testing	Tested system and evaluation results
M6 – Final Submission	Documentation, corrections, and demonstration preparation	Final diploma project

The dates of later milestones will be adjusted according to the approved academic schedule and supervisor feedback.

10.3 Task Management

Development tasks will be recorded in YouTrack. Each task will include a description, priority, responsible person, expected outcome, acceptance criteria, and deadline.

The initial task backlog is presented below.

Table 10.2. Initial task backlog.

Task	Description	Priority	Status
T-01	Prepare project proposal and scope	High	Draft prepared — review pending
T-02	Review scientific literature and existing systems	High	Draft prepared — review pending
T-03	Define functional and non-functional requirements	High	Draft prepared — review pending
T-04	Prepare architecture diagrams	High	Draft prepared — review pending
T-05	Prepare interactive prototype	High	Draft prepared — review pending
T-06	Validate requirements with stakeholders	High	Awaiting consultation
T-07	Design database schema	High	Not started
T-08	Implement KPI management and data entry	High	Not started
T-09	Implement monitoring and reporting	High	Not started
T-10	Perform testing and evaluation	High	Not started

These are planning identifiers, not YouTrack issue IDs. Drafted deliverables are not marked completed until acceptance criteria and supervisor review are satisfied. Actual statuses, assigned YouTrack identifiers, and deadlines will be updated after project setup.

10.4 Version Control and Documentation

The GitHub repository is planned to store source code, documentation, diagrams, prototype materials, configuration examples, and tests. At the time of this draft, GitHub and YouTrack setup and the required contribution history are outstanding.

Development changes will be recorded through descriptive commits linked to YouTrack tasks. Significant changes will be developed in separate branches and submitted through Pull Requests.

The main branch will contain the reviewed project baseline. Selected significant changes will be reviewed with the supervisor at agreed checkpoints.

The first pre-defense submission is planned to use the Git tag pre-defense-1 after the repository is prepared and the submitted materials are checked. No tag is claimed to exist at this stage.

10.5 Milestone Dependencies and Immediate Next Steps

Requirements confirmation must precede final data-model decisions; a testable data model and core functions must precede integration and evaluation. The following dependencies define a realistic development order.

Stage	**Depends on**	**Evidence of completion**
Documentation draft	Approved development strategy and initial research	Draft report, comparison tables and diagrams
Stakeholder clarification	Supervisor/client availability	Consultation notes and approved priorities
Data-model finalization	Confirmed KPI structure and result semantics	Reviewed entity relationships and validation rules
Core system	Data model and selected technology stack	KPI entry, retrieval and basic reports on synthetic data
Verification	Working and testable functions	Reproducible test cases and measured baseline

10.6 Risk Management

Table 10.3. Initial risk register.

ID	**Risk**	**Probability**	**Impact**	**Mitigation Strategy**

R-01	Delays in stakeholder consultation	Medium	High	Prepare initial requirements and arrange consultations early
R-02	Changes in system requirements	High	Medium	Use iterative development and maintain updated documentation
R-03	Limited access to institutional data	Medium	High	Use synthetic data until authorized access is available
R-04	Difficulties integrating selected technologies	Medium	Medium	Evaluate alternatives and test integration early
R-05	Incorrect interpretation of KPI calculations	Medium	High	Define calculation rules for each indicator type and validate them with stakeholders
R-06	Limited development time	High	High	Prioritize essential functions and implement the system incrementally
R-07	Unauthorized access to sensitive information	Medium	High	Apply role-based access control and test permissions

Risk probability and impact are preliminary qualitative judgments, not statistically estimated probabilities. They will be revised as requirements, access permissions, and milestone dates become clearer.

10.7 Progress Monitoring

Project progress will be monitored through YouTrack task statuses, GitHub commits, Pull Requests, and milestone completion.

Each completed task will be checked against its acceptance criteria. Project documentation will be updated when requirements or design decisions change.

Supervisor feedback and corresponding revisions will also be recorded to maintain traceability.

10.8 Conclusion

The proposed project management approach supports structured development, progress tracking, and risk control.

Using GitHub and YouTrack will help maintain a clear record of project activities, technical decisions, and completed deliverables throughout the diploma development process.

11. Supervisor Feedback and Revisions

11.1 Supervisor Consultation

The diploma project is supervised by Aliya Katyetova. The initial project direction was supported through the supervisor's written project justification dated 14 October 2026.

Before the first pre-defense, prepared a draft report, architecture diagrams, an interactive user interface prototype, and a verification and evaluation plan. The project materials were also organized using GitHub for version control and YouTrack for task management.

In October 2026, the supervisor reviewed the project documentation and provided feedback through written communication.

The supervisor stated that the submitted document was well prepared and sufficient for the first pre-defense, provided that it meets the requirements specified in the official Requirements file.

The consultation also clarified the current situation at AITU. According to the supervisor, the university does not currently use an automated system for monitoring the implementation of its Development Strategy. Information collection, implementation-status calculations, and reporting are carried out manually.

The supervisor considered the proposed functionality appropriate for the initial stage of development. However, the final technical solution remains undecided and may involve Power BI or another system.

Therefore, the feedback supports the current project direction while identifying several areas that require further clarification.

11.2 Initial Project Direction and Confirmed Clarifications

Table 11.1. Project direction and supervisor clarifications

Area	Initial Project Direction	Supervisor Feedback / Current Decision
Project relevance	Automation of Development Strategy monitoring	Confirmed that the current monitoring process is manual
Project scope	KPI monitoring, progress tracking, and reporting	Suitable for the initial development stage; further refinement is expected

Requirements	Preliminary functional and non-functional requirements	Must remain consistent with the official Requirements file
System architecture	Proposed logical components and system diagrams	May be revised after the final technical approach is selected
Technology	Preliminary comparison of development and visualization technologies	Power BI or another solution is under consideration; no final decision
Prototype	Interactive demonstration of the proposed monitoring system	Included in the submitted project materials; detailed changes may be discussed later
Institutional requirements	Requirements derived from the AITU Development Strategy	Additional information is expected from a planned technical specification
First pre-defense	Initial report and supporting design materials	Supervisor considers the preparation sufficient, subject to Requirements compliance

The feedback indicates that the project is suitable for continuation but should remain flexible during the next development stages.

In particular, the current architecture, selected technologies, and detailed functional requirements should not be treated as final institutional decisions.

11.3 Feedback and Revision Log

The supervisor's feedback was documented to identify the necessary changes and future development activities.

Table 11.2. Supervisor feedback and revision log

Date	Feedback / Recommendation	Revision or Planned Action	Status

October 2026	No automated monitoring system is currently used at AITU	Update Section 2 to describe the confirmed manual monitoring process	Completed in report draft
October 2026	Information collection, implementation-status calculations, and reporting are performed manually	Include these activities in the problem analysis and project motivation	Completed in report draft
October 2026	Proposed functionality is appropriate for the initial stage	Retain the preliminary system scope while allowing further refinement	Acknowledged
October 2026	Power BI or another solution may be selected	Keep the technology decision provisional and review alternatives during later development	Pending
October 2026	Project materials are sufficient for the first pre-defense if they meet the Requirements file	Verify that all required deliverables and criteria are covered	Verification required
October 2026	A technical specification for digitalizing strategy monitoring may be shared	Review the document when received and update relevant requirements	Awaiting document
October 2026	Information about a possible presentation will be provided later	Prepare presentation materials after receiving official instructions	Awaiting instructions

The revision log distinguishes between changes already reflected in the report and actions that remain planned.

At this stage, updates have been made to the documentation based on the supervisor's feedback. No changes to the software implementation are claimed as completed as a result of this consultation.

11.4 Next Steps

Based on the supervisor's recommendations, the following activities are planned:

1. Verify the first pre-defense report and supporting materials against the official Requirements file.
2. Review the technical specification for digitalizing the AITU strategy monitoring process when it becomes available.
3. Refine the functional requirements and user workflows based on the new information.
4. Reconsider the technical architecture and selected technologies, including the possible use of Power BI or another suitable solution.
5. Update the system prototype, diagrams, and development plan if additional requirements are identified.
6. Continue recording project changes through GitHub and YouTrack.
7. Prepare presentation materials after receiving official instructions for the first pre-defense.

The supervisor's feedback confirms that the initial project direction is appropriate and that the current preparation is sufficient for the first pre-defense, subject to compliance with the official requirements.

At the same time, further consultation and analysis are necessary before finalizing the system architecture and implementation approach.